IMPERIAL COLLEGE LONDON



DISTRIBUTED COMPUTATION AND NETWORKS

Performance Evaluation of Cloud Servers

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INTRODUCTION

In this assignment we us following notation

$$p_i \ldots \text{probability that a message is dispatched to server } i \tag{0.1}$$
$$p(n_i) \ldots \text{probability that server } i \text{ has } n \text{ messages} \tag{0.2}$$
$$p(n_i^+) \ldots \text{probability that server } i \text{ has } n+1 \text{ messages} \tag{0.3}$$
$$p(n_i^-) \ldots \text{probability that server } i \text{ has } n-1 \text{ messages} \tag{0.4}$$
$$r_i \ldots \text{rates of exponentially distributed service time of server } i \tag{0.5}$$

We assume that the Job Dispatcher distributes each job independently of all previous jobs and with the same probability. Moreover we notice that we deal with a feed forward network with independent servers. Hence we can write the external arrival rate λ as sum of arrival rates for each server.

$$\lambda = \sum_{i=1}^{N} a p_i \tag{0.6}$$

1 JOINT PROBABILITY DISTRIBUTION

Due to the absence of negative customers we use following traffic equation.

$$\sum_{i=0}^{N} \Lambda_i P_{ij} = \Lambda_j - \lambda_j \tag{1.1}$$

As all transition probabilities are zero and we have no feedback, it follows

$$0 = \Lambda_j - a p_j \tag{1.2}$$

Hence we obtain that the total arrival rate at each server is it's external arrival rate.

$$\Lambda_j = a p_j \tag{1.3}$$

In order to find the steady state of the system we set the derivative of the change in steady state to zero. Moreover we derive the steady state for each server individually because we can use the property that the servers are decoupled and independent.

$$\frac{dp(n_i)}{dt} = 0 = -(a p_i + r_i 1_{[n_i>0]}) p(n_i) + a p_i p(n_i^-) 1_{[n_i>0]} + r_i p(n_i^+) \tag{1.4}$$

By dividing both sides with $p(n_i)$ and substituting the ratios we obtain

$$a p_i + r_i 1_{[n_i>0]} = a p_i \frac{r_i}{\Lambda_i} 1_{[n(i)>0]} + \frac{\Lambda_i}{r_i} r_i \tag{1.5}$$

$$a p_i = \Lambda_i \tag{1.6}$$

Thus the traffic equation has a unique solution.
We obtain the steady state probability by using Jackson's Theorem. If the traffic equation has a unique solution and each $\Lambda_i < \mu_i$, then $p(k) = \lim_{t\to\infty} p[n_1(t) = n_1, \ldots p_N(t) = n_N]; n = (n_1, \ldots n_N)$ can be written as

$$p(n_1, n_2, \ldots, n_N) = \Pi_{i=1}^{N} p(n_i) = \Pi_{i=1}^{N} \rho_i^{n_i} (1 - \rho_i) \tag{1.7}$$

where $\rho_i = \frac{\Lambda_i}{r_i} = \frac{a p_i}{r_i}$
Side note: In our case we would not need Jackson's Theorem to show that the probabilities are independent as mentioned in Section .

2 Average Response Time

The average queue length can be calculated by following sum for each server, again we use the independence property.

$$\langle J_i \rangle = \frac{\lambda}{\mu} \sum_{n=0}^{\infty} J_i p_0 \left(\frac{a p_i}{r_i}\right)^{n-1} \tag{2.1}$$

$$\langle J_i \rangle = p_0 \frac{a p_i}{r_i} \sum_{n=0}^{\infty} n \left(\frac{a p_i}{r_i}\right)^{n-1} \tag{2.2}$$

using $\rho = \frac{\lambda}{\mu}$ and $\frac{d}{dx} x^n = n x^{n-1}$ we obtain

$$\langle J_i \rangle = p_0 \frac{ap_i}{r_i} \frac{d}{d\rho} \sum_{n=0}^{\infty} \rho^n \tag{2.3}$$

$$\langle J_i \rangle = \frac{\frac{ap_i}{r_i}}{1 - \frac{ap_i}{r_i}} \tag{2.4}$$

Because there is a non zero probability that the systems empties, we can use little's formula to obtain the average queue length per server.

$$\langle W_i \rangle = \frac{\frac{1}{r_i}}{1 - \frac{ap_i}{r_i}} \tag{2.5}$$

Thus the queue length of a server with a low service rate can be reduced by reducing the proportion p_i of messages that are dispatched to it.

The end of this section should include the expression W for average response time of the system.

3 Minimum Average Response Time

The minimization of W is a constraint optimizatin problem. Therefore we attempt to use Lagrange Multipliers. In the following we use the substitution $\alpha_i = \frac{p_i}{r_i}$.

$$W = \sum_{i=1}^{N} \frac{p_i}{r_i(1 - \frac{ap_i}{r_i})} = \sum_{i=1}^{N} \frac{\alpha}{1 - a\alpha}$$

$$s.t. \sum_{i=1}^{N} p_i = 1$$

Formulation for Lagrangian method

$$f(r_1, \ldots, r_N) = \sum_{i=1}^{N} \frac{\alpha}{1 - a\alpha}$$

$$g(r_1, \ldots, r_N) = \sum_{i=1}^{N} \alpha_i r_i - 1$$

$$\mathcal{L}(\alpha_1, \ldots, \alpha_N, \lambda) = \sum_{i=1}^{N} \frac{\alpha}{1 - a\alpha} - \lambda \left(\sum_{i=1}^{N} \alpha_i r_i - 1 \right)$$

The partial derivatives for α_i are

$$\frac{\partial \mathcal{L}(\alpha_1, \ldots, \alpha_N, \lambda)}{\partial \alpha_i} = \frac{1}{(1 - a\alpha_i)^2} - \lambda r_i = 0$$

$$\Rightarrow 1 - a\alpha_i = \frac{1}{\sqrt{\lambda r_i}}$$

$$\Rightarrow \alpha_i = \frac{1}{a}(1 - \frac{1}{\sqrt{\lambda r_i}})$$

The partial derivative for λ is

$$\frac{\partial \mathcal{L}(\alpha_1, \ldots, \alpha_N, \lambda)}{\partial \lambda} = \sum_{i=1}^{N} \alpha_i r_i - 1 = 0$$

$$\Rightarrow \sum_{i=1}^{N} \frac{r_i}{a} \left(1 - \frac{1}{\sqrt{\lambda}\sqrt{r_i}}\right) = 1$$

$$\Rightarrow \lambda = \frac{\sum_{i=1}^{N} \sqrt{r_i}}{\sum_{i=1}^{N} r_i - a}$$

Hence we should choose p_i as following in order to minimize the response time W

$$p_i = \alpha_i r_i = \frac{r_i}{a} \left(1 - \frac{1}{\sqrt{r_i} \frac{\sum_{j=1}^{N} \sqrt{r_j}}{\sum_{j=1}^{N} r_j - a}}\right)$$

$$p_i = \frac{r_i}{a} \left(1 - \frac{\sum_{j=1}^{N} r_j - a}{\sqrt{r_i} \sum_{j=1}^{N} \sqrt{r_j}}\right)$$

Thus the higher the service rate of a server the higher the probability to dispatch a job to that server.

4 Average Queue Length

As derived in Section 2 the average number of jobs $\langle J_i \rangle$ in server i is

$$\frac{a p_i}{r_i} = \frac{a}{r_i} \alpha_i r_i = a \alpha_i = 1 - \frac{\sum_{j=1}^{N} r_j - a}{\sum_{j=1}^{N} \sqrt{r_j} \sqrt{r_i}} \tag{4.1}$$

$$\langle J_i \rangle = \frac{1 - \frac{\sum_{j=1}^{N} r_j - a}{\sum_{j=1}^{N} \sqrt{r_j} \sqrt{r_i}}}{1 - \left(1 - \frac{\sum_{j=1}^{N} r_j - a}{\sum_{j=1}^{N} \sqrt{r_j} \sqrt{r_i}}\right)} \tag{4.2}$$

$$\langle J_i \rangle = \sqrt{r_i} \frac{\sum_{j=1}^{N} \sqrt{r_j}}{\sum_{j=1}^{N} r_j - a} - 1 \tag{4.3}$$

Hence the queue length of a server is a constant weighted with the square root of it's service rate minus 1.

5 Practical Job Dispatching Heuristic

The jobs are quantized and therefore we cannot slice a package according to the best proportion p_i. Therefore we have to find a way to dispatch the jobs in a way that on average we obtain the best proportion p_i. One way to achieve that is by using Weighted Round Robin [1]. In each round it dispatches the jobs according to the proportion p_i. The number of jobs per round is chosen such to to minimize the aberration of the real $\hat{p}_i$ to the desired p_i.

6 Comparison of Queue Memory Setups

In this task we compare a distributed queue as described above and a global queue where the Job Dispatcher sends a job as soon as a server is ready to process a job.
In the second case it is guaranteed that no job is waiting unless all servers are utilized.
We cannot guarantee this in the first case, because we have fluctuations in the processing time due to it's exponential distribution. But the dispatcher only considers the expected processing time. That can cause servers to be idle although there are jobs in the system that are waiting.
Thus it would be better to have a global queue which dispatches a job to a server as soon as it becomes ready to process a new job.

References

[1] M. Katevenis, S. Sidiropoulos, and C. Courcoubetis, "Weighted Round-Robin Cell Multiplexing in a General-Purpose ATM Switch Chip," *IEEE Journal on Selected Areas in Communications*, vol. 9, no. 8, pp. 1265–1279, 1991.